Databases for data analytics

https://github.com/evidencebp/databases-course/

Window Functions

Window Functions Goal

- Usually we look at a single raw record
- Grouping allows us to look on groups of records
- Window functions allows us to look at a single record and “look through the window” on the context

Window definition

- Current record (implicit yet very important)
- Name (in order to reuse it)
- Partition (the relevant grouping)
- Order (sorting the window)
- Frame (context filtering)

Window is a (rather) good metaphor

- Current record - floor
- Name (e.g., left, center, right)
- Partition (**bad metaphor**

  , maybe sides)
- Order (going up or down)
- Frame (which neighboring

  floors can we see)

Given a floor,

specific window functions compute

a value on the floor.

How to use windows functions

Decide what are the anchor rows that you would like to enhance

- They might be computed with views, group bys, etc

Decide on the window context

- Partition - in which column the anchor row is similar to the windows' rows
- Order - what is the timeline, if any, on which the window's rows should be ordered
- Frame - rows filtering by the relative position to the anchor on the timeline
- Name - to improve comprehension and reuse

As in group by, windows may contain many rows.

- Use aggregation and window functions to select a single value
- You can apply many functions to the same window

Common uses

Usually applied to aggregated data in BI systems

- Looking at a month before/after
- Rolling metric (e.g., last 3 month average)
- Specific value (instead of max, the 10th value)
- Percentiles
- Relative position

Resources

- Window functions
- Sales examples
- See IMDB examples

Windows functions implementation has sharp edges and database differences (e.g., count distinct, relations to group by). It is out of the course scope yet remember to use carefully.